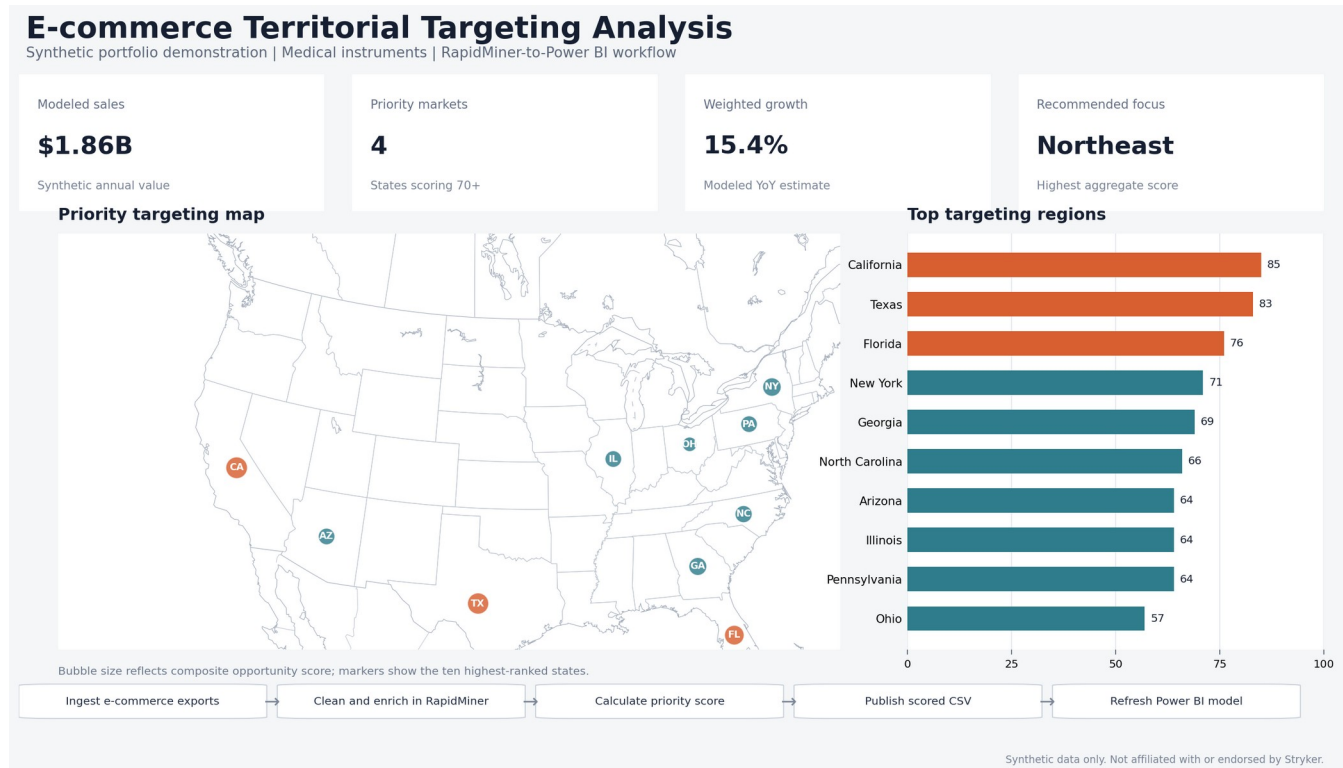


E-COMMERCE TERRITORIAL TARGETING ANALYSIS

Geographic Opportunity Scoring for Medical Instruments

RapidMiner data preparation + Power BI territorial targeting concept



Project statement

Built a synthetic e-commerce targeting model that consolidates commercial and geographic signals, calculates a composite opportunity score, and translates the results into an interactive territorial-planning dashboard.

Independent demonstration only. No confidential Stryker data, actual sales performance, or official territory definitions were used.

1. Business Context and Solution

A practical analytics concept for prioritizing geographic markets and supporting territory design.

Business problem

E-commerce sales can reveal demand that is not fully visible through field activity alone. However, transaction exports, geographic reference data, market-potential indicators, and sales-territory assignments often reside in separate files. The analysis needs a repeatable way to standardize these sources, score opportunity consistently, and surface the result in a map that commercial teams can use.

Project objectives

- Consolidate state- or ZIP-level e-commerce activity into a governed geographic fact table.
- Identify higher-priority targeting regions using sales, order volume, growth, market potential, and current coverage.
- Provide a transparent scoring model that can be audited and adjusted by commercial stakeholders.
- Translate scored outputs into an interactive Power BI design for territory planning and whitespace analysis.

Solution architecture

1	2	3	4	5
Source data	RapidMiner	Scoring model	Published file	Power BI
Orders, revenue, ZIP, product family	Clean, join, aggregate, normalize	Calculate priority and tier	Governed scored CSV or table	Map, ranking, KPIs, drill-through

Portfolio deliverables

Interactive HTML dashboard, editable Word case study, shareable PDF, synthetic Excel data model, and import-ready CSV output.

2. Data Model and Priority Scoring

A transparent composite score converts multiple commercial signals into one targeting rank.

Core analytic grain

The demonstration model uses one row per state. A production implementation would typically move to ZIP5, customer account, or a defined market-basket grain, then roll results into sales territories.

Input	Definition	Weight	Role in score
E-commerce Sales	Synthetic annual revenue	28%	Current commercial value
Online Orders	Synthetic transaction count	20%	Demand volume
YoY Growth	Synthetic annual growth rate	20%	Momentum
Market Potential	Modeled 0-100 opportunity index	20%	Addressable opportunity
Whitespace	One minus modeled coverage	12%	Underpenetrated market potential

Scoring logic

Composite formula

Priority Score = $100 \times [0.28(\text{Sales normalized}) + 0.20(\text{Orders normalized}) + 0.20(\text{Growth normalized}) + 0.20(\text{Market Potential} / 100) + 0.12(\text{Whitespace})]$

Priority tiers and actions

Score band	Tier	Suggested action
72-100	Highest	Prioritize digital campaigns and review field coverage
60-71	High	Develop focused market plan and monitor conversion
45-59	Moderate	Test targeted campaigns and reassess
0-44	Low	Maintain baseline coverage and review quarterly

The weights are intentionally configurable. In a real commercial setting, the model should be calibrated with historical conversion, sales-rep capacity, channel mix, and market-access constraints.

3. Power BI Dashboard Design

The report structure supports both executive prioritization and analyst-level investigation.

Recommended page layout

- Top KPI cards: modeled sales, priority-market count, weighted growth, and recommended focus territory.
- Main map: state or ZIP choropleth colored by Priority Score, with territory boundary overlays where supported.
- Ranking panel: horizontal bar chart of states or markets by Priority Score.
- Detail table: sales, orders, growth, market potential, coverage, whitespace, score, and recommended action.
- Slicers: territory, priority tier, product family, customer segment, and reporting period.

Illustrative DAX measures

Measure	DAX concept
Modeled Sales	SUM(Scored_Data[E-commerce Sales])
Priority Markets	CALCULATE(DISTINCTCOUNT(Scored_Data[State Abbr]), Scored_Data[Priority Score] >= 70)
Weighted Growth	DIVIDE(SUMX(Scored_Data, Scored_Data[YoY Growth] * Scored_Data[E-commerce Sales]), [Modeled Sales])
Whitespace	1 - AVERAGE(Scored_Data[Coverage])

Interaction design

- Selecting a state filters the ranking and detail table while displaying a market-profile tooltip.
- Territory slicers allow leaders to compare portfolio balance and identify overloaded or underdeveloped regions.
- A drill-through page can expose ZIP-level opportunity, target accounts, channel mix, and historical growth.
- The model can support scenario parameters to test alternative weights or coverage assumptions.

Production consideration

For accurate territorial boundaries, use governed ZIP-to-territory assignments and a supported polygon layer such as Azure Maps, ArcGIS for Power BI, or a certified custom shape map. Avoid treating state boundaries as a substitute for actual sales territories.

4. RapidMiner Workflow and Implementation Plan

RapidMiner provides a repeatable preprocessing and scoring layer before the Power BI refresh.

1. Ingest	Read transaction exports and geographic reference files.
2. Validate	Check schema, required fields, data types, nulls, and duplicate keys.
3. Standardize	Format ZIP5 as text, normalize product and segment labels, and derive dates.
4. Enrich	Join geography, territory, market potential, and commercial coverage.
5. Aggregate	Summarize sales and order activity to the selected geographic grain.
6. Normalize	Scale model inputs to comparable 0-1 ranges.
7. Score	Calculate the weighted Priority Score and assign a tier.
8. Publish	Write the scored output to a governed location for Power BI refresh.

Quality controls

- Reject records with missing geographic keys or invalid ZIP formats.
- Reconcile aggregated revenue and order counts back to the source export.
- Track unmatched geography and territory joins in an exception output.
- Version scoring weights and retain run metadata for reproducibility.
- Validate map totals against the scored source table after every refresh.

Portfolio value demonstrated

- Commercial analytics: translated transaction data into a prioritization framework.
- Data engineering: designed a repeatable cleansing, enrichment, aggregation, and publishing workflow.
- Geospatial analysis: connected opportunity signals to state and territory structures.
- Business intelligence: defined an interactive dashboard, measures, filters, and drill paths.
- Governance: documented assumptions, formulas, quality checks, and limitations.

Recommended next iteration

Replace synthetic state data with de-identified ZIP-level transactions, add account and product dimensions, calibrate weights against historical conversion, and validate the recommended territories with field leadership.